

## CO4 - AT2 - CASE STUDY

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## QUESTIONS

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25 marks

4/4 answered

## Q1 CO4 - AT2 - CASE STUDY

25 marks

A large enterprise resource planning application is built from multiple source files, including modules for sales, inventory, and finance, and all of these modules must reference a common variable that stores the current fiscal year. The variable is defined once in one source file but must be read and, in some cases, modified by functions compiled separately in other source files. Without proper declaration, each module would either fail to compile or maintain its own inconsistent copy of the fiscal year. The design must ensure a single, consistent value is shared across the entire application.

Question: Identify which storage class allows the fiscal-year variable to be accessed across multiple source files, and explain the role of extern in achieving this shared access.

## Your Code

```
1 #include <stdio.h>
2 int fiscalYear = 2025;
3
4 void updateFiscalYear(int newYear) {
5     fiscalYear = newYear;
6     printf("Finance Module: Fiscal year updated to %d\n", fiscalYear);
7 }
8 extern int fiscalYear;
9
10 void showSalesReport() {
11     printf("Sales Module: Current fiscal year is %d\n", fiscalYear);
12 }
13 extern int fiscalYear;
14
```

## Input

stdin input

## Output

Q4  
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## Your Code

```
1 #include <stdio.h>
2 int fiscalYear = 2025;
3
4 void updateFiscalYear(int newYear) {
5     fiscalYear = newYear;
6     printf("Finance Module: Fiscal year updated to %d\n", fiscalYear);
7 }
8 extern int fiscalYear;
9
10 void showSalesReport() {
11     printf("Sales Module: Current fiscal year is %d\n", fiscalYear);
12 }
13 extern int fiscalYear;
14
15 void showInventoryStatus() {
16     printf("Inventory Module: Current fiscal year is %d\n", fiscalYear);
17 }
18
19 int main() {
20     printf("---- Initial Status ----\n");
21     showSalesReport();
22     showInventoryStatus();
23
24     printf("\n---- Finance updates fiscal year ----\n");
25     updateFiscalYear(2026);
26
27     printf("\n---- Status After Update ----\n");
28     showSalesReport();
29     showInventoryStatus();
30
31     return 0;
32 }
```

## Input

stdin input

## Output

```
---- Initial Status ----
Sales Module: Current fiscal year
is 2025
Inventory Module: Current fiscal
year is 2025
```



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Your Code

```

1 #include <stdio.h>
2 #include <string.h>
3 typedef enum {
4     ELECTRICITY,
5     WATER,
6     GAS
7 } UtilityType;
8
9 struct SmartMeterRecord {
10     int meterID;
11     long timestamp;
12     char location[30];
13     UtilityType type;
14
15     union {
16         float electricityKWh;
17         float waterLiters;
18         float gasCubicMeters;
19     } reading;
20 };
21
22 int main() {
23
24     struct SmartMeterRecord record1;
25     record1.meterID = 101;
26     record1.timestamp = 1672531200;
27     strcpy(record1.location, "Block A, Apt 4");
28     record1.type = ELECTRICITY;
29     record1.reading.electricityKWh = 12.5;
30
31     struct SmartMeterRecord record2;

```

Input

stdin input

Output

```

Meter ID: 101 | Location: Block
A, Apt 4
Reading: 12.50 kWh

Meter ID: 102 | Location: Block
A, Apt 4

```



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Your Code

```

1 #include <stdio.h>
2 #include <string.h>
3
4 union ClaimDetails {
5     char hospitalDetails[50];
6     char accidentDetails[50];
7     char propertyDamage[50];
8 };
9
10 struct Claim {
11     int claimID;
12     char policyholderName[50];
13     char claimDate[15];
14     char status[20];
15     int claimType;
16     union ClaimDetails details;
17 };
18
19
20 void updateClaimStatus(struct Claim *ptr, const char *newStatus) {
21     if (ptr != NULL) {
22         strcpy(ptr->status, newStatus);
23         printf("Claim ID %d status updated to: %s\n", ptr->claimID, ptr->status);
24     }
25 }
26
27 int main() {
28
29     struct Claim claim1;
30     claim1.claimID = 101;
31     strcpy(claim1.policyholderName, "Kaviyaasri M");
32     strcpy(claim1.claimDate, "2023-10-27");

```

Input

stdin input

Output

```

Initial Claim: ID 101, Holder:
Kaviyaasri M, Status: Pending,
Info: City General Hospital
Claim ID 101 status updated to:
Approved
Updated Claim Status: Approved

```

25 marks

Q4  
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25 marks

4/4 answered

Your Code

```
1 #include <stdio.h>
2
3 #define MAX_PRODUCTS 100
4
5 typedef struct {
6     int productCode;
7     int quantity;
8     int supplierID;
9 } InventoryItem;
10
11 int main() {
12
13     InventoryItem inventory[MAX_PRODUCTS];
14
15     inventory[0].productCode = 101;
16     inventory[0].quantity = 50;
17     inventory[0].supplierID = 5001;
18
19     inventory[1].productCode = 102;
20     inventory[1].quantity = 30;
21     inventory[1].supplierID = 5002;
22
23     int targetCode = 101;
24     int newShipment = 25;
25
26     InventoryItem *ptr = NULL;
27
28     for (int i = 0; i < 2; i++) {
29         if (inventory[i].productCode == targetCode) {
30             ptr = &inventory[i];
31             break;
```

Input

stdin input

Output

Product Code: 101  
Updated Quantity: 75